Faculty of Engineering

Module Title: Engineering Chemistry

Sheet 2: (Model Answer)

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1. Choose the correct answer:

A. ……of thermodynamics is used to understand the concept of energy conservation

- Zeroth law
- **First law**
- Second law

B. In thermodynamics ……is not a state variable

- Density
- Entropy
- **Heat**

C. A well-stoppered thermos flask contains some ice cubes. This is an example of……

- Closed system
- Open system
- **Isolated system**

D. The least random state of the water system is:

- **Ice**
- Liquid water
- Steam

E. The first law of thermodynamics states that energy cannot be

- Created only
- Destroyed only
- Converted
- **Created and destroyed**

F. Zeroth's law of thermodynamics helped in the creation of which scale?

- **Temperature**
- Heat energy
- Pressure
- Internal energy

G. A system……..

- **Is a specified region, where a transfer of energy and mass are to be studied**
- Cannot transfer either energy or mass to or from the surroundings
- Is a region of constant mass and only energy is allowed to cross the boundaries
- Has mass transfer across its boundaries and the mass within the system is not constant

H. An open system is one in which

- Neither heat nor mass crosses the boundary of the system
- The mass of the working substance crosses the boundary of the system, but the heat does not cross the boundary of the system
- **Both the heat and mass of the working substance crosses the boundary of the system**

2. Which of the following processes are spontaneous? Which are nonspontaneous?

(a) Diffusion of perfume molecules from one side of a room to the other

(b) Heat flows from a cold object to a hot object

Answer:

a) Spontaneous

b) Nonspontaneous

3. Predict the sign of ΔS in the system for each of the following processes:

(a) $CO_{2(s)} \rightarrow CO_{2(g)}$ (sublimation of dry ice)

(b) $CaSO_{4(s)} \rightarrow CaO_{(s)} + SO_{3(g)}$

(c) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

Answer:

a) The molecules in a gas are free to move about randomly, whereas the molecules in a solid are tightly held in a highly ordered arrangement. Therefore, randomness increases when a solid sublimes and ΔS is positive.

b) One mole of gaseous molecules appears on the product side of the equation, and none appears on the reactant side. Because the reaction increases the number of gaseous molecules, the entropy change ΔS is positive.

c) The entropy change ΔS is negative because the reaction decreases the number of gaseous molecules from 4 mol to 2 mol. Fewer particles can move independently after the reaction than before.

4. Internal energy increases by 150J while 50J of heat is supplied. How much work is done by the gas?

Answer:

$$\Delta U = Q - W$$

$$W = Q - \Delta U = 50J - 150J = -100J$$

5.A system receives 150 calories of heat and work done by the system is 418 J.

What is change in its internal energy?

Answer:

$$\Delta U = Q - W = (150)(4.186) - (418) = 209\ J$$